

Herald



CLAUDE.md

Field	Value
-------	-------

Revision	13
----------	----

Created	2026-05-15
---------	------------

Last modified	2026-05-27
---------------	------------

Status	active
--------	--------

Status summary	
----------------	--

r13: propagated HelixConstitution §11.4.95–§11.4.97 (workable-items SQLite DB is tracked-in-git not gitignored, safe-parallel-work-with-long-build catalogue + mandate, maximum-use-of-idle-time + progress-update cadence) as short-form restatements citing the literal anchors 11.4.95/11.4.96/11.4.97, inherited per §11.4.35; restated + cited, not redefined. Prior r12: propagated HelixConstitution §11.4.89–§11.4.94 (background-test execution, Obsolete status + obsolescence audit, summary-doc clarity, multi-pass change-evaluation, SQLite workable-items SSoT, zero-idle parallel-by-default) as short-form restatements citing the literal anchors, inherited per §11.4.35; restated + cited, not redefined.** Prior r11: propagated HelixConstitution §11.4.85 (stress + chaos test mandate) + §11.4.87 (endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing) + §11.4.88 (background-push: commit-lock release immediately after commit, push detached) into this file's covenant cluster — short-form restatements citing the literal section numbers, inherited per §11.4.35; required by the §11.4.87 CM-COVENANT-114-87-PROPAGATION pre-build gate which asserts the literal anchor 11.4.87 is present in every consuming repo's CLAUDE.md/AGENTS.md/QWEN.md. Herald restates + cites, does not redefine or weaken. Prior r10: Wave 6 (pherald inbound runtime + Claude Code headless bridge — code-doc closure; tag v0.4.0 deferred to T13b post-live-evidence).** New pherald listen Cobra subcommand drives the closed-loop runtime: Telegram getUpdates long-poll + bot self-filter (§32.9 anti-echo via bot.Me.Username) + OnPhoto/OnDocument/OnVoice handlers with sha256 content-addressed attachment download (~/.herald/inbox/<sha>.<ext>, idempotent) + Claude Code dispatch with Opus pinned in argv (claude --model claude-opus-4-7, §33.7) + envelope pre-text — verbatim operator wording "We have received new message from our communication channel ..." preceding the existing <<<HERALD-DISPATCH-v1>>> block (§33.6) + <<<HERALD-REPLY>>> parser with action routing (reply / issue.open / event.emit) + tgram.SendReply wiring reply_to_message_id. T10a --qa-out-dir JSONL journaling (§107.x). 12 Wave-6 commits (T1=ad87d7f → T12=96c7c6b). 8 new e2e invariants

Field	Value
	E63-E70 (currently SKIP-with-documented-reason — convert to PASS once T10b lands operator-supplied live evidence). Wave 6 mutation gate tests/ test_wave6_mutation_meta.sh (3 paired hermetic mutations, 4/4 PASS). HRD-100 atomic Issues→Fixed. Spec V3 r11→r12 with §32.9 + §33.6 + §33.7 + §43 pherald listen row. Workspace at 16 modules (Wave 5 added qaherald; Wave 6 added NO new module — inbound is pherald/ internal/inbound/). Prior r9: Wave 5 T1 qaherald module skeleton landed (16th workspace module + 8th flavor binary). Prior r8: Wave 4b TOON status pointer; tag v0.3.0 substrate. Prior r5/r6: §107 End-user-usability covenant section restating the verbatim operator mandate at Herald level + ToC entry; ties to HERALD_CONSTITUTION.md §107 + inheritance-gate invariant I8a (paired with I8b/c on AGENTS.md + HERALD_CONSTITUTION.md).
Issues	none
Issues summary	—
Fixed	spec-path references (r2), pre-implementation-language update (r3), submodules + HRD-docs + codegraph-index enumeration (r4), §107 mandate restatement + I8a anchor (r5), Wave-2/3a workspace-module-count refresh (r7), Wave 4a + 4b status pointer + e2e/mutation tally refresh (r8), Wave 5 T1 qaherald skeleton + commons_tls enumeration fold-in (r9), Wave 6 inbound runtime + CC headless bridge code-doc closure (r10), Helix §11.4.85 + §11.4.87 + §11.4.88 propagation (r11), Helix §11.4.89–§11.4.94 propagation (r12), Helix §11.4.95–§11.4.97 propagation (r13)
Fixed summary	aligned with HRD-009/HRD-009b/HRD-013/HRD-014 landing in the same commit; r4 closes the discoverability gaps observed during a fresh /init review; r5 closes the Herald-level explicit-restatement gap identified by the 2026-05-20 audit; r7 closes the doc drift observed during the Wave 3a final review; r8 records the Wave 4a→4b transport/wire-format substrate upgrades; r9 lands Wave 5 T1 (qaherald/cmd/qaherald + commons/branding.go qa flavor) and back-fills the Wave 4a commons_tls workspace enumeration that was missing from r7/r8; r10 lands Wave 6 — pherald inbound runtime (pherald listen Cobra subcommand, pherald/internal/inbound/ Dispatcher with <<<HERALD-REPLY>>> action routing, tgram bot self-filter + OnPhoto/ OnDocument/OnVoice with sha256 content-addressed attachment download, claude_code dispatcher pinned to Opus with verbatim pre-text envelope, tgram.SendReply with reply_to_message_id, T10a --qa-out-dir JSONL journaling); spec V3 r11→r12 + HRD-100 atomic Issues→Fixed; tag v0.4.0 deferred to T13b (post-T10b live evidence); r11 propagates the three new inherited HelixConstitution mandates (§11.4.85 stress+chaos test mandate, §11.4.87 endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing, §11.4.88 background-push) into this file's covenant cluster as short-form restatements per the §11.4.87 CM-COVENANT-114-87-PROPAGATION pre-build gate (literal anchors 11.4.85/11.4.87/11.4.88 now present), inherited per §11.4.35; restated + cited, not redefined; r12 propagates the next six inherited HelixConstitution mandates (§11.4.89 background-test execution, §11.4.90 Obsolete status + obsolescence audit, §11.4.91 summary-doc clarity, §11.4.92 multi-pass change-evaluation, §11.4.93 SQLite workable-items SSoT, §11.4.94 zero-idle parallel-by-default) into the same covenant cluster as short-form restatements citing the literal anchors 11.4.89/11.4.90/11.4.91/11.4.92/11.4.93/11.4.94, inherited per §11.4.35; restated + cited, not redefined; r13 propagates the next three inherited HelixConstitution mandates (§11.4.95 workable-items SQLite DB tracked-in-git-not-gitignored, §11.4.96 safe-parallel-

Field	Value
Continuation	work-with-long-build catalogue + mandate, §11.4.97 maximum-use-of-idle-time + progress-update cadence) into the same covenant cluster as short-form restatements citing the literal anchors 11.4.95/11.4.96/11.4.97, inherited per §11.4.35; restated + cited, not redefined. bump again when T10b lands operator-supplied live evidence under docs/qa/HRD-100-<run-id>/ and T13b tags v0.4.0; then continue with Wave 7 (genericize messenger-channel framework per operator mandate 2026-05-22 — Slack/Max/Email next), Wave 3c carry-over (HRD-024 iherald paging, HRD-033 destructive-guard body, remaining HRD-018..028 constitution bindings), §43 command catalogue HRD-029..056, comprehensive docs audit (task #147), and the OOM-Protect Herald flavor (oherald, future post-Wave-6 closure per operator 2026-05-22).

Table of contents

- [INHERITED FROM Helix Constitution \(parent-discovery\)](#)
- [Project status](#)
- [End-user-usability covenant \(Herald §107 / Helix §11.4 — MANDATORY ANTI-BLUFF\)](#)
- [Inherited covenant restatements — Helix §11.4.85 / §11.4.87 / §11.4.88 / §11.4.89 / §11.4.90 / §11.4.91 / §11.4.92 / §11.4.93 / §11.4.94 / §11.4.95 / §11.4.96 / §11.4.97](#)
- [Mission \(from the spec\)](#)
- [Intended stack](#)
- [Multi-host mirror convention](#)
- [Inheritance gate \(run before any commit touching root docs\)](#)
- [Spec-change rule \(load-bearing — docs/specs/mvp/specification.V3.md §"Specification documents"\)](#)
- [Project conventions from the spec \(apply when scaffolding\)](#)
- [constitutable/ directory \(parent-project extension hook\)](#)
- [Documentation artefacts \(PDF/HTML siblings\)](#)
- [Notes for future scaffolding](#)

This file provides guidance to Claude Code (claude.ai/code) when working with code in this repository.

INHERITED FROM Helix Constitution (parent-discovery)

Herald is consumed as a submodule of a parent project that already carries the Helix Constitution submodule at <parent>/constitution/. Herald therefore does **NOT** keep its own copy. Locate the constitution from any nested depth by walking up parents until you find <ancestor>/constitution/Constitution.md, or by running the canonical helper:

```
CONST_DIR="$(bash "$(find . -type d -name constitution -print
    -quit 2>/dev/null)/find_constitution.sh)"
# or, more robustly, from any starting directory:
CONST_DIR="$(bash <ancestor>/constitution/find_constitution.sh)"
```

For standalone development of Herald (no parent project), clone the constitution alongside Herald:

```
git clone git@github.com:HelixDevelopment/HelixConstitution.git \
    $(dirname "$PWD")/constitution
```

Once located, all rules in `<discovered>/CLAUDE.md` and the `<discovered>/Constitution.md` it references apply unconditionally. Herald-specific rules below extend them — they MUST NOT weaken any inherited rule. When this file disagrees with the discovered constitution, the constitution wins.

Canonical: <https://github.com/HelixDevelopment/HelixConstitution>

Read order on a cold start:

1. `<discovered-constitution>/CLAUDE.md + Constitution.md` — universal Helix rules. Inherited unconditionally.
2. `<discovered-constitution>/AGENTS.md` — agent guardrails (anti-bluff, no-guessing, paired mutations).
3. `README.md` — Herald overview + how-to.
4. This file (Herald-specific notes below).
5. `docs/guides/HERALD_CONSTITUTION.md` — Herald's project-specific constitutional extensions.
6. `docs/guides/CONSTITUTION_INHERITANCE.md` — operator/agent guide for the discovery contract + the inheritance gate.
7. `docs/specs/mvp/specification.V3.md` — mission spec (mostly TBD).

Project status

Herald is in **first-implementation cycle (r1)** as of 2026-05-20. The Go scaffold has landed; live-integration work (HRD-010..HRD-012, HRD-016) is still open. The repo contains:

- `README.md` — mission, deployment model, inheritance contract, quickstart.
- `docs/specs/mvp/specification.V3.md` — MVP spec stub (substantive sections TBD).
- `docs/guides/HERALD_CONSTITUTION.md` — Herald's project constitution (extends Helix).
- `docs/guides/CONSTITUTION_INHERITANCE.md` — operator/agent guide for parent-discovery + gate semantics.
- `upstreams/` — Herald's mirror declarations (see below).
- `tests/test_constitution_inheritance.sh + _meta.sh` — paired inheritance gate (§1.1).
- `.gitignore` tuned for Go + `.DS_Store`.

Herald does **not** ship a `constitution/` submodule of its own; the parent project provides it. See `docs/guides/CONSTITUTION_INHERITANCE.md`.

As of 2026-05-20 the Go scaffold has landed (first-implementation cycle r1). The repo now contains:

- `go.work` (gitignored locally; check in if the project wants reproducible workspaces — current convention: gitignored per spec §9.1).
- `commons/ (L0)` — `commons/types.go` with the full §11.0 Channel contract + `Subscriber/CloudEventEnvelope/TraceContext/Branding/ChannelID`; `commons/clock.go` Clock abstraction; `commons/uuidv7.go`; `commons/branding.go` per-flavor factory.
- `commons_prefix/` — §8.2 3-letter prefix algorithm.
- `commons_messaging/channels/null/` — full §11.14 `null://` sandbox adapter (working, tested).

- `commons_messaging/channels/tgram/` — Telegram adapter SCAFFOLD (HRD-011 open).
- `commons_messaging/dispatch/claude_code/` — Claude Code session-resolution + envelope formatter; live `claude --resume` invocation pending (HRD-012 open).
- `commons_storage/` — 9 SQL migrations (000001..000005) embedded via `//go:embed`; `pgx + River + Redis` wiring pending (HRD-010 open).
- `pherald/cmd/pherald/` — Cobra CLI; `pherald version` works end-to-end; other subcommands stubbed with HRD-NNN error pointers.
- `quickstart/` — Herald-specific Docker/Podman Compose + Dockerfile + `otel-config + .env.example` per §26.5 (HRD-008 open for operator validation). Migrated from `containers/quickstart/` 2026-05-20 when the `containers/` submodule was added.
- `containers/` — git submodule pointing at `git@github.com:vasic-digital/containers.git` (the `digital.vasic.containers` Go module — runtime auto-detection + on-demand container boot + lifecycle/health). Imported by Foundation tests + the `pherald doctor` subcommand to bring Postgres + Redis + OTel up on-demand.

Build + test: from the repo root:

```
go build ./commons/... ./commons_prefix/... ./
    commons_messaging/... ./commons_storage/... ./
    commons_constitution/... ./commons_infra/... ./
    pherald/...
go test -race -count=1 ./commons/... ./commons_prefix/... ./
    commons_messaging/... ./commons_storage/... ./
    commons_constitution/... ./commons_infra/... ./
    pherald/...
```

Tests pass on Go 1.25+ (verified on 1.26). Workspace is configured via `go.work` listing **16** Herald modules: the 7 foundation modules (`commons`, `commons_constitution`, `commons_infra`, `commons_messaging`, `commons_prefix`, `commons_storage`, `pherald`) plus `commons_auth` (Wave 3a JWT verifier / Gin middleware), `commons_tls` (Wave 4a auto-cert loader for the dual HTTP/3+TLS listener), and the 7 flavor binaries: the 6 Wave 2 flavors (`sherald`, `cherald`, `bherald`, `rherald`, `iherald`, `scherald`) plus `qaherald` (Wave 5 — QA bot, `pherald` [?](#) Telegram round-trip automation per §107.x docs/qa evidence mandate). `go.work` itself is gitignored per spec §9.1; a fresh clone needs `go work init && go work use ./...` (or copy the existing snippet from the project's local working tree). The 9 Helix-stack submodules under `submodules/` are referenced via `replace` directives in the consuming modules' `go.mod`, not via `go.work`.

Anti-bluff verification (run before any release tag or risky commit):

```
scripts/audit_antibluff.sh      # 3 invariants: §11.4 anchor +
    tests + paired meta
scripts/codegraph_validate.sh   # CodeGraph index integrity (7
    probes)
scripts/e2e_bluff_hunt.sh       # 14 end-to-end checks against
    real services
```

CodeGraph index lives in `.codegraph/` (`codegraph.db + config.json`, both gitignored). Rebuild with `scripts/codegraph_setup.sh` when source layout changes; `codegraph_validate.sh` will FAIL otherwise.

`scripts/e2e_bluff_hunt.sh` is the canonical end-to-end smoke per Universal §11.4. It builds pherald, runs the full test suite, starts a real Gin server + hits every `/v1` route + asserts response bodies, boots a real Postgres container via the `containers/` submodule, runs the M2 integration tests against it, and graceful-shutdowns. ALL 14 invariants must PASS — a single FAIL means a real feature is broken for end users. At 2026-05-20 it reports 14/14 PASS.

When the user asks to "add a feature" the spec is the source of truth — find the relevant §, then the relevant module + package, then the relevant HRD-NNN if one is already open. New work opens a new HRD-NNN in `docs/Issues.md` per V3 §8.3 lifecycle.

Do not invent build/test commands beyond what `go test ./<module>/...` provides. Live-integration tests (Telegram bot, Claude Code session, real Postgres) require operator-supplied credentials — see `docs/CONTINUATION.md` for the live-test handoff prompt.

End-user-usability covenant (Herald §107 / Helix §11.4 — MANDATORY ANTI-BLUFF)

Forensic anchor — verbatim operator mandate:

"all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product! This MUST BE part of Constitution of our project, its CLAUDE.MD and AGENTS.MD if it is not there already, and to be applied to all Submodules's Constitution, CLAUDE.MD and AGENTS.MD as well (if not there already)!"

The bar for shipping any Herald feature is **NOT** "tests pass" — it is **"the end user of the flavor binary can actually use the feature."** Every PASS (unit, integration, gate, Challenge, smoke, e2e) MUST carry positive runtime evidence that the user-visible behaviour works. Metadata-only / configuration-only / "absence-of-error" / grep-only PASS are §11.4 PASS-bluffs and constitute critical defects regardless of how green the summary line looks. Canonical Herald authority: `docs/guides/HERALD_CONSTITUTION.md` §107. Canonical Helix authority: `<discovered>/Constitution.md` §11.4 + §11.4.1..§11.4.16 and `<discovered>/CLAUDE.md` "MANDATORY ANTI-BLUFF COVENANT — END-USER QUALITY GUARANTEE". Canonical Herald evidence: `scripts/e2e_bluff_hunt.sh` (14 invariants against real services; ALL must PASS). Inheritance gate invariant **I8a** asserts this covenant anchor is present in this file.

§107.x — docs/qa/ Evidence Mandate (operator mandate, 2026-05-22; cascades from Helix §11.4.83)

Forensic anchor — verbatim operator mandate (2026-05-22):

"every feature that ships MUST carry a recorded e2e communication transcript + any attached materials under `docs/qa/<run-id>/` (per-feature subdirectories). A feature with no QA transcript is itself a §107 PASS-bluff — it claims to work but has no auditable runtime evidence. Bot-driven automation (e.g. Herald's planned `qaherald` binary) MUST preserve full bidirectional communication threads as proof."

Every Herald feature that ships — every flavor binary (pherald, sherald, cherald, bherald, rherald, iherald, scherald, future qaherald) and every /v1/* route they expose — MUST carry a recorded end-to-end communication transcript plus all attached materials (Telegram screenshots, Gin response bodies in JSON or TOON, OpenTelemetry trace exports, container logs) committed under docs/qa/<run-id>. A Herald feature that ships without a docs/qa/<run-id> directory is by definition a §107 PASS-bluff — its e2e_bluff_hunt PASS line claims it works for end users, but no auditable runtime evidence proves an end user ever exercised it.

Operative rule for Herald. (1) Every Herald HRD-NNN (V3 §8.3) work-item that introduces a user-visible feature MUST land its docs/qa/HRD-NNN/ directory (timestamp-prefixed if multiple runs) in the same logical work effort. (2) Bidirectional transcripts only — for Telegram-driven features (HRD-011, planned qaherald), capture both directions of the bot conversation; for Gin/v1 routes, capture both the request payload and the full response body. (3) Attached materials commit verbatim — never link to external Slack/Drive/Telegram URLs; the artefact lives in-repo. (4) The planned qaherald binary (HRD-NNN to be assigned) is Herald's QA bot: it drives pherald [2] Telegram round-trips and preserves the full conversation thread under docs/qa/qaherald-<TS>/. (5) Release gates (scripts/release.sh when implemented + the existing tag-time guard) REFUSE to tag a release whose feature-shipping commits lack their docs/qa/<run-id>/. (6) scripts/e2e_bluff_hunt.sh invariants for new features MUST cite the docs/qa/<run-id>/ artefact as their positive-evidence anchor (§11.4.2 / §11.4.5 composition).

Cascade authority. Helix Universal Constitution §11.4.83 (the verbatim operator mandate is anchored there). Herald §107.x is the project-binding restatement.

Enforcement. A feature commit that lacks docs/qa/<run-id>/ triggers FAIL at the release-gate layer. The §11.4.4 test-interrupt-on-discovery applies — the entire release cycle stops until evidence lands.

Non-compliance is a release blocker. No --qa-evidence-optional, --qa-transcript-later, --qa-bot-summary-suffices flag exists for Herald.

§107.y — Working-Tree Quiescence Rule (operator mandate, 2026-05-22; cascades from Helix §11.4.84)

Short tag: working-tree quiescence.

Forensic anchor — verbatim operator mandate (2026-05-22):

"no subagent commit may proceed while any concurrent mutation gate is in flight in the same checkout. Before git add, the committing agent MUST grep its own working tree for mutation markers (MUTATED for paired, // always pass, return json.Marshal shortcut paths, etc.). Any unexplained file in the staging area triggers ABORT."

Lesson (forensic case study — Herald-internal). On 2026-05-21 a logo-fix subagent (commit 72e81ab, "Fix: replace pandoc {width=96px} image attr with HTML tag") ran in this very checkout while a paired §1.1 Wave 4b mutation gate had temporarily injected an // always pass shortcut into commons_auth/middleware.go (JWT-bypass mutation, intended for the paired-mutation cycle to mutate → assert FAIL → restore). The logo-fix subagent's git add swept the mutation residue into its commit; the resulting commit was pushed to all four mirrors before any other agent caught it. Within the hour the SECURITY FIX (d5bd360,

"SECURITY FIX: restore commons_auth/middleware.go JWT verify (mutation residue in 72e81ab)") restored the verify path, but the production-equivalent-binary-with-bypassed-JWT window is a real security-defect window — small, but non-zero, and demonstrably exploitable during that interval. This is no longer hypothetical; it happened. The rule below is the constitutional outcome.

Operative rule for Herald.

1. **Pre-git add quiescence check.** Every commit flow (main thread + subagent) MUST grep the working tree for the canonical Herald mutation markers BEFORE git add:
 - MUTATED for paired (the canonical paired-§1.1 marker emitted by tests/test_wave4b_mutation_meta.sh)
 - // always pass, // MUTATION, # MUTATION (Go + shell mutation annotations)
 - return json.Marshal shortcut paths in commons or commons_messaging (Wave 4b TOON mutation residue)
 - _mutated_* filename suffixes
 - .git/MUTATION_IN_PROGRESS (the lockfile — see (3))
2. **Scope-match.** Cross-check git status --porcelain against the subagent's declared scope. Any file outside the declared scope → ABORT. The subagent MUST explicitly account for every modified / untracked / staged entry.
3. **Lockfile serialisation.** When any mutation gate is in flight, the gate's first action is touch .git/MUTATION_IN_PROGRESS; its last action (in trap-on-exit) is rm .git/MUTATION_IN_PROGRESS. Any subagent that finds this lockfile present MUST refuse to git add and ABORT until the gate completes the mutate → assert-FAIL → restore → assert-PASS cycle and removes the lockfile.
4. **Worktree isolation (preferred).** When parallel subagents are required (§11.4.20 / §11.4.70 subagent-driven default), prefer git worktree add per subagent over single-checkout concurrency — eliminates the cross-mutation race by construction.
5. **Pre-push mutation-residue scanner.** scripts/mutation_residue_audit.sh (to be implemented; HRD-NNN to be assigned) MUST run before every push. Any commit in the pushed range containing a mutation marker → push BLOCKED, commit MUST be reverted or amended before mirrors are updated.

Prototype enforcement. tests/test_wave4b_mutation_meta.sh ALREADY includes a working-tree quiescence check (check_quiescence() at line 92; assertion at line 197 — "Working-tree quiescence — assert no MUTATED markers leaked"). It is the canonical prototype. Generalising it across all paired-§1.1 gates is open work; the universal scanner (scripts/mutation_residue_audit.sh) is the planned roll-out.

Cascade authority. Helix Universal Constitution §11.4.84. Herald §107.y is the project-binding restatement.

Enforcement. A mutation marker that lands in any tagged Herald commit is a critical defect regardless of how briefly it persisted — see commit 72e81ab / d5bd360 as proof.

Non-compliance is a release blocker. No --allow-residue, --skip-quiescence, --mutation-cleanup-later flag exists.

Inherited covenant restatements — Helix §11.4.85 / §11.4.87 / §11.4.88 / §11.4.89 / §11.4.90 / §11.4.91 / §11.4.92 / §11.4.93 / §11.4.94 / §11.4.95 / §11.4.96 / §11.4.97 (inherited per §11.4.35)

These twelve mandates are **inherited** from the HelixConstitution via parent-discovery (§11.4.35). Herald **restates** + **cites** them here — it does **NOT** redefine or weaken them. The literal anchors below are required by the §11.4.87 CM-COVENANT-114-87-PROPAGATION pre-build gate, which asserts that 11.4.85, 11.4.87, and 11.4.88 appear in every consuming repo's CLAUDE.md / AGENTS.md / QWEN.md.

§11.4.85 — Stress + Chaos Test Mandate (Helix, 2026-05-24)

Every fix or improvement landed in Herald MUST ship with full-automation **stress** AND **chaos** test suites — sustained/concurrent load + failure-injection (process-death, network-fault, input-corruption, resource-exhaustion, state-corruption) — each PASS citing a captured-evidence artefact per §11.4.5 + §11.4.69. A fix that PASSES its happy-path test but has never been exercised under stress or fault-injection is a §11.4 / §107 PASS-bluff at the resilience layer. For Herald this binds the flavor binaries (pherald listen inbound long-poll under concurrent updates, Gin /v1/* routes under sustained load, claude_code dispatch under process-death, container flows under disk/OOM pressure) — stress + chaos evidence lands under docs/qa/<run-id>/stress_chaos/ and is cited by the matching e2e_bluff_hunt invariant. Canonical authority: HelixConstitution Constitution.md §11.4.85 (inherited per §11.4.35).

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing (Helix, 2026-05-26)

When the operator instructs Herald work to "continue in endless loop fully autonomously" (or equivalent), it is a HARD-CONTRACT covenant: (A) continue until ALL are simultaneously TRUE — Herald's autonomous loop checks docs/Issues.md Status-column has zero In progress/Ready for testing/In testing/Reopened entries, docs/CONTINUATION.md §3 "Active work" is empty, the TaskList reports no subagent mid-execution, and no in-flight external dependency (build, push, sync) remains; (B) dispatch background subagents for parallelisable, non-contending work rather than serialising — idle is permitted ONLY while waiting on a result; (C) every closed item lands four-layer test coverage (§11.4.4(b)) with real captured-evidence PASS; (D) anti-bluff end-to-end — tests AND Challenges are bound equally; (E) the loop terminates only on all-clear, an explicit operator STOP, a §12 host-session-safety demand, or a scheduled wake against a known-future-actionable signal. No --idle-OK / --skip-endless-loop / --metadata-only-test-suffices escape hatch exists. Canonical authority: HelixConstitution Constitution.md §11.4.87 (inherited per §11.4.35).

§11.4.88 — Background-push mandate: commit-lock release immediately after commit, push runs detached (Helix, 2026-05-26)

Herald commit flows MUST release the commit-lock (.git/.commit_all.lock) the instant git commit returns 0 — BEFORE any push runs — because the commit is durable on local disk regardless of push outcome; gating further local work on a remote round-trip is the exact zero-idle anti-pattern §11.4.87 prohibits. The push then runs **detached** (nohup ./push_all.sh ... & then disown), with per-remote flock (.git/.push.<remote>.lock) serialising same-remote invocations while DIFFERENT

remotes (GitHub / GitLab / GitFlic / GitVerse) push in parallel. Backgrounded push failures land in `qa-results/push_failures/<TS>_<remote>.log` and the next autonomous-loop tick MUST surface them (silent push-failure is a §11.4 PASS-bluff at the distribution layer). The ONLY synchronous-push escape is the explicit `--sync-push` flag for §11.4.41 force-push merge-first paths. Canonical authority: HelixConstitution Constitution.md §11.4.88 (inherited per §11.4.35).

§11.4.89 — Background test execution mandate (Helix, 2026-05-27)

Any Herald test cycle expected to exceed ~30s — the `tests/test_wave*_mutation_meta.sh` gates, `scripts/e2e_bluff_hunt.sh`, future stress/chaos suites — MUST run detached (`nohup ... > qa-results/<test_id>_<TS>.log 2>&1 & + disown`); the main work stream returns immediately to the §11.4.42 priority queue and polls the log/exit-status rather than blocking. Composes with §107.y / §11.4.84: a backgrounded mutation gate still mutates the shared tree, so only ONE runs against the main checkout at a time (serialised by the `.git/MUTATION_IN_PROGRESS` lockfile + `scripts/mutation_residue_audit.sh` pre-push scanner); concurrent gates require separate `git worktree` checkouts. Foreground permitted only for <30s tests or on explicit operator request. Canonical authority: HelixConstitution Constitution.md §11.4.89 (inherited per §11.4.35).

§11.4.90 — Obsolete status + per-item obsolescence audit (Helix, 2026-05-27)

Herald's HRD Status closed-set gains a 4th terminal value `Obsolete` ($\rightarrow$ `Fixed.md`) for items no longer valid (Reason closed-set: superseded-by-design-change / superseded-by-later-mandate / feature-removed / duplicate-of / unsupported-topology). Every `Obsolete` HRD heading MUST carry an `**Obsolete-Details:**` line within 8 non-blank lines: Since (ISO date), Reason, Superseding-item (§/HRD ref), Triple-check evidence (git-log/grep/runtime path per §11.4.6 — bare assertion forbidden). At every release-gate sweep, re-evaluate every open + Fixed HRD for obsolescence; migrations are atomic per §11.4.19. Canonical authority: HelixConstitution Constitution.md §11.4.90 (inherited per §11.4.35).

§11.4.91 — Summary-doc clarity (Helix, 2026-05-27)

Every one-liner in `docs/Issues_Summary.md` / `docs/Fixed_Summary.md` / `docs/Status_Summary.md` / `README` doc-link rows MUST be a self-contained clause (≥ 6 words OR ≥ 40 chars) naming the SUBJECT + the PROBLEM/GOAL — never a section-label fragment (Composes with, Closure criteria), bare metadata (Critical, Bug), a status restatement, or a bare HRD-id. Derive each from the source long-form H1/H2 heading, never invented. Canonical authority: HelixConstitution Constitution.md §11.4.91 (inherited per §11.4.35).

§11.4.92 — Multi-pass change-evaluation discipline (Helix, 2026-05-27)

Every non-trivial Herald change MUST pass — and document — a 5-pass evaluation before it is commit-ready: Pass 1 main-task captured-evidence (§11.4.5, no "should work"); Pass 2 regression-blast-radius (every touched file + every importer/caller audited); Pass 3 cross-feature interaction (shared state/timing/env — e.g. a gate edit checks §107.y quiescence + §11.4.89 backgrounding); Pass 4 deep-research validation (§11.4.8 — external precedent or literal "NO external solution found"); Pass 5 anti-bluff confirmation (no metadata-only/config-only/script-bug PASS). Evidence lands in the commit footer or `docs/qa//qa-results/`. Trivial changes (typo, revision-bump, MD-export regen touching zero source) are exempt only with explicit

commit-message citation. Canonical authority: HelixConstitution Constitution.md §11.4.92 (inherited per §11.4.35).

§11.4.93 — SQLite-backed single-source-of-truth for workable items (Helix, 2026-05-27)

Herald's text-based HRD trackers (`docs/Issues.md / Fixed.md / *_Summary.md / Status.md / CONTINUATION.md` §3) migrate to a SQLite single-source-of-truth (`docs/.workable_items.db`, **version-controlled as Herald's authoritative SSoT** per operator mandate 2026-05-27 — a deliberate Herald divergence from the parent §11.4.93 gitignored-with-regeneration default, since for Herald the DB IS the authoritative artefact and **MUST** be committed, not regenerated-on-clone; only the transient WAL/SHM sidecars stay ignored) with bidirectional MD[?]DB regeneration so sync-drift is mechanically impossible. The Go binary lives in the constitution submodule (`constitution/scripts/workable-items/`) — Herald references it per §11.4.74 catalogue-first, never reimplements. Migration is 6-phase; Herald files the tracking HRD (Phase 1) and progresses incrementally. Canonical authority: HelixConstitution Constitution.md §11.4.93 (inherited per §11.4.35).

§11.4.94 — Zero-idle priority-first parallel-by-default operating mode (Helix, 2026-05-27)

Binding always-on contract: Herald work is **NEVER** idle while a priority-queued item can progress. Before any wake/sleep/"waiting for X", survey the priority queue, identify all non-contending items, and dispatch them in parallel (subagent-driven per §11.4.20/§11.4.70 when non-trivial; background per §11.4.89 when >30s). Pick highest-Severity/priority first. The conductor remains the integration + commit + push seam; parallel work **MUST NOT** compromise stability (composes with §107.y quiescence + §11.4.92 multi-pass + §12 host-safety). Idle is permitted **ONLY** when every item is genuinely blocked on external dependency, the operator issued STOP, or §12 host-session-safety demands it. Canonical authority: HelixConstitution Constitution.md §11.4.94 (inherited per §11.4.35).

§11.4.95 — Workable-items SQLite DB is TRACKED in git, never gitignored (Helix, 2026-05-27)

Herald **ALREADY** complies (operator correction 2026-05-27, recorded in §108.h + `.gitignore`): the workable-items SQLite DB is version-controlled, committed + pushed alongside every state change, WAL-checkpointed (`PRAGMA wal_checkpoint(TRUNCATE)`) before commit so only the transient `-wal/-shm` sidecars stay gitignored, and never force-rewritten without a §9.2 hardlinked-backup. This is an explicit §11.4.30 carve-out and **AMENDS** the earlier §11.4.93 / §108.h "gitignored-with-regeneration" text. **Herald alignment note:** the constitution's canonical path is `docs/workable_items.db` (no leading dot); Herald's HRD-131 currently references `docs/.workable_items.db` — reconcile to the canonical path when the DB is implemented (HRD-131 Phase 2+, currently deferred). Canonical authority: HelixConstitution Constitution.md §11.4.95 (inherited per §11.4.35).

§11.4.96 — Safe-parallel-work-with-long-build catalogue + mandate (Helix, 2026-05-27)

Herald has no AOSP build, but the principle binds: during ANY long-running Herald operation (a backgrounded mutation gate, `scripts/e2e_bluff_hunt.sh`, a §11.4.85 stress/chaos suite, a doc export) the conductor **MUST** consult the safe-parallel catalogue and dispatch non-

contending work rather than idle. SAFE-in-parallel for Herald: (A) MD/docs work, (B) `scripts/` helpers, (C) gate authoring, (D) test authoring, (E/F) commit + push to mirrors, (H) read-only analysis subagents, (I) web research, (J) workable-items DB ops. UNSAFE-during-a-running-gate (maps to §107.y): `git checkout / reset --hard / clean` on files a gate is transiently mutating, a SECOND concurrent mutation gate against the same checkout, host-session-safety breaches (§12). Subagent-driven default per §11.4.20 / §11.4.70. Canonical authority: HelixConstitution Constitution.md §11.4.96 (inherited per §11.4.35).

§11.4.97 — Maximum-use-of-idle-time mandate + progress-update cadence (Helix, 2026-05-27)

(A) Every minute of conductor idle time during which progressable, non-externally-blocked work exists is a violation — dispatch work continuously through the whole idle window, not just at scheduled wakes. (B) Emit concise (1–3 line) operator-facing progress updates at milestone boundaries with NO prompt required: every HEAD advance (what landed), every subagent return (integrated), every constitutional anchor propagated, every captured-evidence artefact (`docs/qa/ / qa-results/` path), every Issues→Fixed/Obsolete closure. (C) Continuous physical-proof gathering per §11.4.5 / §11.4.6 / §11.4.69 — every closed item carries positive captured evidence committed alongside the closure. (E) Idle ONLY when genuinely blocked (operator STOP, external dependency, §12 host-safety). Canonical authority: HelixConstitution Constitution.md §11.4.97 (inherited per §11.4.35).

Mission (from the spec)

Ingesting system events and reliably fanning them out to multiple notification channels so every alert reaches the right destination without confusion.

The spec mentions "Integration into the Constitution" as a planned section — Herald is intended to plug into a larger governance/policy system that lives outside this repo. Treat that as a real constraint when designing interfaces, even though the details are TBD.

Intended stack

Go, inferred from `.gitignore (*.test, go.work, go.work.sum, coverage.*, profile.cov)`. When you scaffold, default to standard `cmd/ + internal/` layout unless the user asks otherwise, and use `go test ./... / go build ./... / go vet ./... —` there is no project-specific tooling to override these yet.

Multi-host mirror convention

`upstreams/` contains one script per mirror host:

- `GitHub.sh` — primary: `git@github.com:vasic-digital/Herald.git`
- `GitLab.sh` — `git@gitlab.com:vasic-digital/herald.git`
- `GitFlic.sh` — `git@gitflic.ru:vasic-digital/herald.git`
- `GitVerse.sh` — `git@gitverse.ru:vasic-digital/Herald.git`

Each script exports a single `UPSTREAMABLE_REPOSITORY` variable and is meant to be **sourced** (`. upstreams/GitHub.sh`), not executed for its output. The naming is intentional — capitalization matches the host's brand (GitFlic, GitVerse). Don't normalize these to lowercase or collapse them into one file; the per-host split is the interface other tooling (likely external mirror-push scripts) keys on.

When adding a new mirror, copy an existing script and change only the URL — keep the `#!/bin/bash` shebang, blank line, and `export` form identical so any consumer that greps these files keeps working.

Inheritance gate (run before any commit touching root docs)

```
bash tests/test_constitution_inheritance.sh
      # 6 invariants (I1-I6), 9 checks
bash tests/test_constitution_inheritance_meta.sh # paired §1.1
      mutation proof
```

The gate inline-walks parents for `<ancestor>/constitution/Constitution.md`. I5 is split into I5a–d (one check per root doc that must declare parent-discovery: `CLAUDE.md`, `AGENTS.md`, `docs/guides/HERALD_CONSTITUTION.md`, `README.md`). I6 forbids re-introducing `<repo-root>/constitution/` or `.gitmodules` — the §104 invariant.

If either script fails, fix at root cause per Universal §11.4.4. Never silently accept the FAIL.

Spec-change rule (load-bearing — docs/specs/mvp/specification.V3.md §"Specification documents")

Any modification to a file under `docs/specs/` (any depth) triggers **mandatory comprehensive planning and implementation** of the implied changes — you may not edit the spec in isolation. This rule does **not** apply to creating or renaming files; for those, ask the operator what to do with the new path. Treat every spec edit as a project-wide ripple, not a doc tweak.

Project conventions from the spec (apply when scaffolding)

These are declared in `docs/specs/mvp/specification.V3.md` and are easy to miss because no code enforces them yet:

- **Workable-item prefix:** HRD- (e.g. HRD-001). Use it for issues, status entries, fix logs.
- **Flavor binaries:** each Herald flavor ships as its own CLI binary, named `<prefix>herald` — `pherald` (Project Herald), `sherald` (System Herald), etc. Designed for CI / pipeline / cron / AI-agent invocation.
- **Layered shared code:** `commons` → `commons_messaging` (level 1) → ... → flavor. Put new shared code in the **lowest layer that still makes sense**; flavors inherit upward. `commons_messaging` owns the Telegram / Max / Slack / Email / Markdown-export integrations.
- **Messenger integration priority:** Telegram → Max → Slack (then Email, then Markdown/PDF/HTML export). Microsoft Teams, Lark, Discord, WhatsApp, Viber are explicitly later iterations — don't pre-implement.
- **Conversation diary:** every message in/out is appended to `docs/herald/diary/main.md` and re-exported to `main.pdf` + `main.html` in sync. Don't break this invariant when designing channel I/O.
- **Container stack:** Postgres (main DB) + Redis (in-memory) bundled via the `containers` submodule (<https://github.com/vasic-digital/containers>). All container names start with `herald`; all host ports start with 70XXX (70001, 70002, ...) to avoid collisions.

- **Credentials:** `.env` (git-ignored) with a committed `.env.example`. Resolution order: exported shell vars from `.bashrc/.zshrc` load first; `.env` is fallback only (never overrides shell exports). **For step-by-step setup of every supported messenger + LLM dispatcher**, see `docs/guides/OPERATOR_CREDENTIALS.md` — comprehensive guide covering Postgres, Redis, Telegram (HRD-011), Claude Code (HRD-012), plus reserved env-var names for planned channels (Slack, Email, Max, Teams, ...) and dispatchers. The guide includes an audit checklist to run before every commit.
- **Vendored SDKs:** any official/unofficial messenger SDK or API client we depend on goes in as a **git submodule**, e.g. `commons_messaging/sdk/telegram` or `commons_messaging/api/telegram` — not go get'd into `go.mod`.

constitutable/ directory (parent-project extension hook)

The empty `constitutable/` directory at the repo root is intentional. Per the spec, a parent project may drop additional `Constitution.md` / `CLAUDE.md` / `AGENTS.md` (in `constitutable/`, `constitutable/<flavor>/`, `constitutable/<flavor>/<variant>/`, etc.) to layer extensions or overrides on top of the discovered `constitution/` submodule. Apply-order: `constitution/` submodule → `constitutable/` extensions → Herald's own docs. Do not delete the directory because it's empty.

Documentation artefacts (PDF/HTML siblings)

`docs/guides/HERALD_CONSTITUTION.md` and `docs/guides/CONSTITUTION_INHERITANCE.md` each ship with a committed `.pdf` sibling. When you edit one of these Markdown files, the PDF goes stale — flag it; do not regenerate silently unless the operator asks.

The HRD-lifecycle docs in `docs/` also ship as PDF/HTML/DOCX quadruples: `Issues.md` (open HRDs per V3 §8.3), `Fixed.md` (closed-HRD log per §11.4.19 atomic migration), `Status.md` (status summary), `CONTINUATION.md` (live-test handoff prompt for operator-supplied credentials). The `*_Summary.md` variants are derived; do not hand-edit.

Logo branding (added 2026-05-21). Every tracked Markdown doc now leads with a centered Herald logo header (pandoc-friendly `<div align="center">` wrapping a `` image reference to `assets/logo/herald_logo_square_128.png`). The export pipeline propagates that logo into the HTML, PDF, and DOCX siblings:

- Logo source: `assets/logo/herald_logo.png` (1664x928 RGB master). Square + transparent variants live under `assets/logo/herald_logo_square_{32,64,128,256,512,1024}.png` (chroma-keyed white → alpha). `assets/logo/herald_logo.svg` is a vector wrapper around the 512px PNG. `assets/logo/print.css` carries print/screen styling for the HTML/PDF route.
- Injection: idempotent re-runnable via `python3 scripts/branding_inject_logo.py <md ...>` — skips submodules/, containers/, `constitutable/`, `docs/diary/`, `LICENSE`; respects YAML front-matter; computes the relative path per doc depth.
- Export: `bash scripts/export_docs.sh [<md>...]` regenerates HTML (pandoc), PDF (`--pdf-engine=weasyprint`), DOCX (pandoc-native) for every `.md`

that already has at least one sibling artefact committed. Pass no args to regenerate everything.

- When you add a new `.md`, run the injector once; when you edit one whose sibling exports are tracked, run the exporter for that one file.

Notes for future scaffolding

- `submodules/` holds 9 vendored Helix-stack modules (each its own `git@github.com:vasic-digital/<name>.git` repo): `auth`, `background`, `cache`, `config`, `database`, `eventbus`, `middleware`, `observability`, `recovery`. They are referenced via `replace` directives in the consuming Herald modules' `go.mod`, NOT via `go.work` (which lists the **14** Herald-owned modules — 7 foundation + `commons_auth` + 6 flavor binaries). Do not add the Helix-stack submodules to `go.work`.
- The repo is in `main` branch and committed under "Milos Vasic" — no other contributors yet.
- `.claude/` exists but is empty; project-local Claude config can go there.
- `LICENSE` is present (do not overwrite without asking).
- `.DS_Store` is now git-ignored; do not re-add the previously-stray files.